DANIEL LAW EN JIE

STUDENT IN ROBOTICS AND MECHATRONICS ENGINEERING

CONTACT

+61433256373

dlaw0014@student.monash.edu

2305/33 Mackenzie St,
Melbourne VIC 3000

https://www.linkedin.com/in/daniel-law-930a6130b/

TECHNICAL SKILLS

CAD/Mechanical Design

- SolidWorks, Onshape, Fusion
- Mechanical assemblies
- Tolerance & fit design

Manufacturing

- FDM 3D Printing – PLA, PETG
- CO_2 Laser Cutting
- Fibre Laser Engraving
- CNC / Milling / Drilling

Electronics

- PCB Assembly
- Soldering
- Sensor Integration

Programming/Robotics

- Python, C/C++, MATLAB
- PLC / CODESYS
- Computer Vision
- Autonomous Navigation & Path Planning

SOFT SKILLS

- Reliable under pressure
- Strong project management skills
- Friendly and positive attitude
- Problem solving

LANGUAGES

- English (Fluent)
- Mandarin (Fluent)
- Malay (Intermediate)
- Cantonese (Intermediate)

PROFILE

Mechatronics and Robotics Engineering student with hands-on experience in CAD design, mechanical product development, prototyping and manufacturing. Interested in expanding across all areas of mechatronics, particularly electronics, robotics and product development, while gaining broader hands-on engineering experience.

ENGINEERING EXPERIENCE

Unique Micro Design

Mechatronics Engineer (Casual) — Jan 2026 - Present
Engineering Intern — Nov 2025 - Jan 2026

- Designed and manufactured **custom 3D-printed enclosures** and mechanical parts, incorporating electronics and connectors constraints.
- **Worked closely with clients** from concept through prototyping and production, through design reviews, feedback and iterative prototyping.
- Gained hands-on experience in PCB assembly, soldering, CO_2 laser cutting, fibre laser engraving and jig design.

PROJECTS/STUDENT TEAM

Monash Connected Autonomous Vehicle (MCAV)

Mechanical Team Member — Apr 2026 – Present

- Designed and manufactured a 4WD, 4-wheel-steer RC vehicle, including chassis, drivetrain, suspension, wheels and Ackermann steering.
- Collaborated within a multidisciplinary mechanical, electrical and software team.

Autonomous Mapping and Object Detection Robot — 2025

- Implemented ArUco localisation, Kalman filtering, YOLOv8 object detection and Dijkstra path planning to map and navigate through objects.

Warman Design and Build Competition — 2025

- Designed, built and programmed an autonomous robot to collect and deposit load while navigating rough terrain.

ACHIEVEMENTS & QUALIFICATIONS

Warman Design and Build Competition (Monash Malaysia) - Champion (1st Place) — 2025

Certified SolidWorks Associate (CSWA) — 2025

Monash High Achiever Award - Bachelor of Engineering — 2023

EDUCATION

Bachelor of Robotics and Mechatronics Engineering — 2023-Present
School of Engineering | Monash University
GPA: 3.88 / 4.0 **WAM:** 87.591

Monash University Foundation Year (MUFY) — 2022-2023
Monash University
MUFY Score: 96.25